

Probability, Conditional Probability, Prior, Posterior and Likelihood

Foundations of Reasoning under Uncertainty

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September 16, 2026

Outline

- 1 From Search to Probability
- 2 Basics of Probability
- 3 Conditional Probability
- 4 Joint Probability
- 5 Bayes' Theorem
- 6 Bayes Classifier & Conditional Independence

Why Shift from Search to Probability?

- Classical search algorithms (BFS, DFS, A*, Minimax) assume:

- A **fully observable, deterministic** environment
- A known set of actions with **certain outcomes**

⇒ Great for puzzles, games, route planning.

- But the **real world** is messy:

- Sensors are noisy (a robot's camera, a self-driving car's lidar)
- Outcomes of actions are uncertain (will it rain tomorrow?)
- We rarely have *complete* information

⇒ We need a mathematical language for **reasoning under uncertainty**

⇒ That language is **Probability Theory**

Search → Probability: A Paradigm Shift

Search-based AI

- State is known exactly
- Transitions are deterministic
- Goal: find a path / plan
- Example: Chess, 8-puzzle, Maze

Probabilistic AI

- State is a *belief* (a distribution)
- Transitions/observations are noisy
- Goal: estimate likely states, make decisions
- Example: Spam filters, Diagnosis, Self-driving cars

Search: "What is true?"



Probability: "What is *likely* true?"

The Turing Award Connection

Probabilistic Reasoning in AI

Judea Pearl — ACM A.M. Turing Award **2011** (awarded 2012)

“For fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning.”

- Pioneered **Bayesian Networks**
- Formalized **conditional independence** using graphs
- Laid the foundation for modern probabilistic AI

Compare: Neural Networks / Deep Learning

Geoffrey Hinton, Yann LeCun, Yoshua Bengio — ACM A.M. Turing Award **2018**

For conceptual and engineering breakthroughs that made deep neural networks a critical component of computing.

*Two Turing Awards, two pillars of modern AI: **probabilistic reasoning** and **deep learning***

Recap: Basic Probability

Sample Space & Events

- Sample space Ω : set of all possible outcomes
- Event $A \subseteq \Omega$
- $0 \leq P(A) \leq 1$, $P(\Omega) = 1$

Basic Rules

- Complement: $P(\bar{A}) = 1 - P(A)$
- Addition (union): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- If A, B are mutually exclusive: $P(A \cup B) = P(A) + P(B)$

Conditional Probability

Definition

The probability of event A **given** that event B has occurred:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

- Read as: “probability of A given B ”
- Knowing B happened *updates* our belief about A
- Rearranged, this gives the **product rule**:

$$P(A \cap B) = P(A | B) P(B) = P(B | A) P(A)$$

Quick Check: Conditional Probability

Problem 1

In a class of **60** students:

- 36 students like AI
- 24 students like ML
- 18 students like **both** AI and ML

If a randomly chosen student likes AI, what is the probability that they also like ML, i.e., find $P(\text{ML} \mid \text{AI})$?

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Solution

$$P(\text{ML} \mid \text{AI}) = \frac{P(\text{ML} \cap \text{AI})}{P(\text{AI})} = \frac{18/60}{36/60} = \frac{18}{36} = 0.5$$

Definition

$P(A \cap B)$, also written $P(A, B)$, is the probability that **both** A and B happen together.

- Captures the full picture of how two variables co-occur
- Marginal probability is recovered by **summing out** the other variable:

$$P(A) = \sum_b P(A, B = b) \quad (\text{sum rule / marginalization})$$

- Building block for joint distributions with more than two variables

Visualizing Joint Probability: The Probability Box

A “joint probability box” for two binary variables A, B :

	B	$\bar{B}$	
A	$P(A, B)$	$P(A, \bar{B})$	$P(A)$
$\bar{A}$	$P(\bar{A}, B)$	$P(\bar{A}, \bar{B})$	$P(\bar{A})$
	$P(B)$	$P(\bar{B})$	

Quick Check: Joint Probability

Problem 2

Two binary variables: **Rain** and **Traffic**. Joint probabilities:

	Traffic = Yes	Traffic = No
Rain = Yes	0.15	0.05
Rain = No	0.10	0.70

Find: (a) $P(\text{Rain} = \text{Yes})$ (b) $P(\text{Traffic} = \text{Yes})$ (c) $P(\text{Rain} = \text{Yes} \mid \text{Traffic} = \text{Yes})$

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Solution

$$P(\text{Rain} = \text{Yes}) = 0.15 + 0.05 = 0.20$$

$$P(\text{Traffic} = \text{Yes}) = 0.15 + 0.10 = 0.25$$

$$P(\text{Rain} = \text{Yes} \mid \text{Traffic} = \text{Yes}) = \frac{0.15}{0.25} = 0.6$$

Deriving Bayes' Theorem

From the product rule, both of the following hold:

$$P(A \cap B) = P(A \mid B) P(B) \quad P(A \cap B) = P(B \mid A) P(A)$$

Equating the right-hand sides and rearranging:

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

- This is **Bayes' Theorem**
- Lets us **invert** a conditional probability: from $P(B \mid A)$ we can get $P(A \mid B)$
- Named after Rev. Thomas Bayes; formalized and popularized in AI by **Judea Pearl**

Prior, Likelihood, and Posterior

$$\underbrace{P(A | B)}_{\text{Posterior}} = \frac{\overbrace{P(B | A)}^{\text{Likelihood}} \overbrace{P(A)}^{\text{Prior}}}{\underbrace{P(B)}_{\text{Evidence}}}$$

- **Prior** $P(A)$: what we believe about A *before* seeing evidence
- **Likelihood** $P(B | A)$: how probable the evidence B is, *if* A were true
- **Posterior** $P(A | B)$: updated belief about A *after* seeing evidence B
- **Evidence** $P(B)$: total probability of observing B (normalizing constant)

Key Idea

Posterior $\propto$ Likelihood $\times$ Prior — this is how an AI agent **updates its beliefs** as new evidence arrives.

Quick Check: Bayes' Theorem

Problem 3 (Medical Diagnosis)

A rare disease affects **1%** of a population. A test for it is **99%** sensitive (true positive rate) and gives a **5%** false positive rate. If a person tests positive, what is $P(\text{Disease} \mid +)$?

Quick Check: Bayes' Theorem

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Solution

$$P(D) = 0.01, P(+ \mid D) = 0.99, P(+ \mid \bar{D}) = 0.05, P(\bar{D}) = 0.99$$

$$P(+) = P(+ \mid D)P(D) + P(+ \mid \bar{D})P(\bar{D}) = (0.99)(0.01) + (0.05)(0.99) = 0.0594$$

$$P(D \mid +) = \frac{(0.99)(0.01)}{0.0594} \approx 0.1667$$

Surprising? Even with a positive test, the probability of disease is only $\approx 16.7\%$ — priors matter!

Conditional Independence

Definition

A and B are **conditionally independent** given C if:

$$P(A, B \mid C) = P(A \mid C) P(B \mid C)$$

Equivalently: $P(A \mid B, C) = P(A \mid C)$

- Once C is known, learning B gives **no extra information** about A
- This assumption is what makes many real-world probabilistic models **computationally tractable**
- Core idea behind **Bayesian Networks** (Judea Pearl) and the **Naive Bayes Classifier**

The Bayes Classifier

- Goal: given features $x_1, x_2, \dots, x_n$, predict class C
- By Bayes' theorem:

$$P(C \mid x_1, \dots, x_n) = \frac{P(x_1, \dots, x_n \mid C) P(C)}{P(x_1, \dots, x_n)}$$

- Modeling the joint likelihood $P(x_1, \dots, x_n \mid C)$ directly is hard (too many combinations!)

Naive Bayes Assumption

Assume all features are **conditionally independent** given the class C :

$$P(x_1, \dots, x_n \mid C) = \prod_{i=1}^n P(x_i \mid C)$$

$$\Rightarrow P(C \mid x_1, \dots, x_n) \propto P(C) \prod_{i=1}^n P(x_i \mid C)$$

Summary

- Real-world AI needs **probabilistic reasoning**, not just search
- Judea Pearl (Turing Award 2011) formalized this with Bayesian networks
- **Conditional probability**: $P(A | B) = P(A \cap B) / P(B)$
- **Joint probability**: full picture of two variables together
- **Bayes' Theorem**: $\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$
- **Conditional independence** makes Bayesian models tractable
- **Naive Bayes Classifier**: applies Bayes' theorem + conditional independence

Questions?